

Ex.No.6 - 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objective – 1: Analyze Different Subcarrier Spacing Values

MATLAB Program

```
clc;

clear;

close all;

scs = [15 30 60 120 240];

mu = 0:4;

bar(scs)

grid on

set(gca,'XTick',1:5)

set(gca,'XTickLabel',mu)

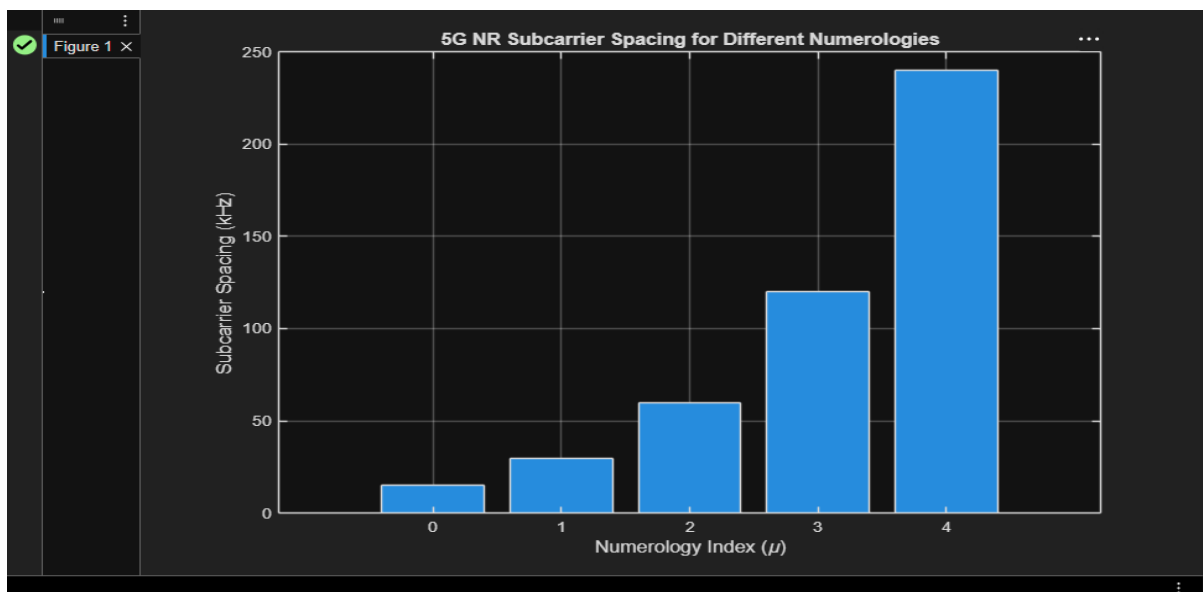
% Subcarrier Spacing (kHz)

% Numerology Index

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

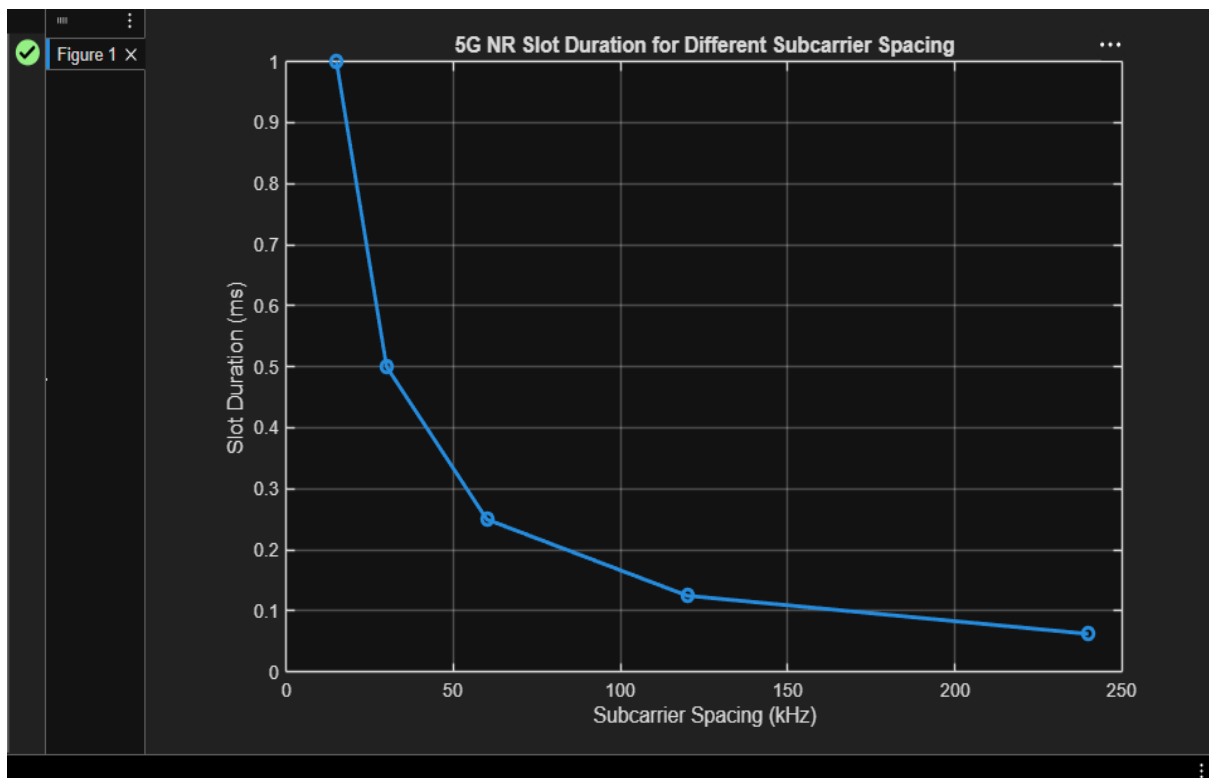
title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objective – 2 Compare Slot Duration

MATLAB Program

```
clear;  
  
close all;  
  
scs = [15 30 60 120 240];  
  
% Subcarrier Spacing (kHz)  
  
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)  
  
plot(scs, slot, 'o-', 'LineWidth', 2)  
  
grid on  
  
xlabel('Subcarrier Spacing (kHz)')  
  
ylabel('Slot Duration (ms)')  
  
title('5G NR Slot Duration for Different Subcarrier Spacing')
```



Objective – 3 Study the Number of OFDM Symbols

MATLAB Program

```
clc;  
clear;  
close all;  
scs = [15 30 60 120 240];  
symbols = [14 14 14 14 14];  
figure  
bar(scs, symbols)  
% Subcarrier Spacing (kHz)  
% OFDM Symbols per Slot  
grid on  
xlabel('Subcarrier Spacing (kHz)')  
ylabel('Number of OFDM Symbols')  
title('OFDM Symbols per Slot for Different Subcarrier Spacing')  
ylim([0 16])
```

